

Homework Assignment 3: Criticality in the S_N Approximation

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The method

All three questions solve the same monoenergetic, steady-state problem,

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi = \frac{1}{2} \left[\Sigma_s \phi + \frac{1}{k} \nu \Sigma_f \phi \right], \quad \phi(x) = \int_{-1}^1 \psi(x, \mu) d\mu \quad (1)$$

with the same three ingredients: a Gauss–Legendre quadrature in angle, diamond differencing in space with a negative-flux fixup, and the Bell & Glasstone k iteration. Lengths are mean free paths in Questions 3 and 4 and centimetres in Question 5.

Angle. The N ordinates $\{\mu_m\}$ and weights $\{w_m\}$ are the Gauss–Legendre nodes on $[-1, 1]$, so that $\mu_{N-m+1} = -\mu_m$, $\sum_m w_m = 2$, and

$$\phi_i = \sum_{m=1}^N w_m \psi_{i,m} \quad (2)$$

Space, slab. Integrating over cell i and closing with the diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i+1/2,m} + \psi_{i-1/2,m})$ gives the sweep

$$\psi_{i,m} = \frac{2|\mu_m| \psi_{\text{in},m} + \frac{1}{2} S_i \Delta x}{\Sigma_t \Delta x + 2|\mu_m|}, \quad \psi_{\text{out},m} = 2\psi_{i,m} - \psi_{\text{in},m} \quad (3)$$

swept from $x = a/2$ inwards for $\mu_m < 0$ and from $x = 0$ outwards for $\mu_m > 0$.

Space, sphere. The spherical equation carries an angular redistribution term,

$$\mu \frac{\partial \psi}{\partial r} + \frac{1 - \mu^2}{r} \frac{\partial \psi}{\partial \mu} + \Sigma_t \psi = \frac{S}{2} \quad (4)$$

whose conservative form integrates over the cell to

$$\mu_m \left(A_{i+\frac{1}{2}} \psi_{i+\frac{1}{2},m} - A_{i-\frac{1}{2}} \psi_{i-\frac{1}{2},m} \right) + \frac{A_{i+\frac{1}{2}} - A_{i-\frac{1}{2}}}{w_m} \left(\alpha_{m+\frac{1}{2}} \psi_{i,m+\frac{1}{2}} - \alpha_{m-\frac{1}{2}} \psi_{i,m-\frac{1}{2}} \right) + \Sigma_t V_i \psi_{i,m} = \frac{S_i}{2} V_i \quad (5)$$

with $A = 4\pi r^2$ and $V_i = \frac{4\pi}{3}(r_{i+1/2}^3 - r_{i-1/2}^3)$. This is (3) with a second outgoing face, in angle, closed by the same diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i,m+1/2} + \psi_{i,m-1/2})$. The angular coefficients follow from demanding that a flat flux solve (5) exactly:

$$\alpha_{\frac{1}{2}} = 0, \quad \alpha_{m+\frac{1}{2}} = \alpha_{m-\frac{1}{2}} - w_m \mu_m, \quad \alpha_{N+\frac{1}{2}} = 0 \quad \text{since} \quad \sum_m w_m \mu_m = 0 \quad (6)$$

The recursion needs a starting value $\psi_{i,1/2}$ at $\mu_{1/2} = -1$, where $1 - \mu^2 = 0$ and the angular derivative drops out; what remains, $-\partial\psi/\partial r + \Sigma_t\psi = S/2$, is the plain slab sweep at $|\mu| = 1$, run inwards from the vacuum boundary.

Fixup. Whenever a diamond outgoing flux $2\psi_{i,m} - \psi_{in,m}$ comes out negative it is set to zero and the cell balance is re-solved with it held there, in angle as well as in space. This never occurs in Questions 3–5: the fission source is proportional to the flux itself, so no cell is ever starved, and a critical system is thin enough that $\Sigma_t\Delta x/|\mu_m| < 2$ at every ordinate.

Boundary conditions. Reflective at the symmetry point, $\psi_{1/2,m} = \psi_{1/2,m'}$ with $\mu_{m'} = -\mu_m$; vacuum at the outer face, $\psi_{I+1/2,m} = 0$ for $\mu_m < 0$.

Eigenvalue. Outer iteration l fixes the fission source $Q_{f,i}^{(l)} = \nu\Sigma_{f,i}\phi_i^{(l)}/k^{(l)}$; inner iteration n sweeps $S_i = \Sigma_{s,i}\phi_i^{(n)} + Q_{f,i}^{(l)}$ until ϕ settles; then

$$k^{(l+1)} = k^{(l)} \frac{P^{(l+1)}}{P^{(l)}}, \quad P^{(l)} = \sum_i \nu\Sigma_{f,i}\phi_i^{(l)} V_i \quad (7)$$

Dividing the fission source by k makes a solution exist at *every* system size, so criticality becomes a root search: `brentq` on $k(\text{size}) - 1$, which costs 8 evaluations of k .

The meaning of c . Questions 3 and 4 specify only $c = (\Sigma_s + \nu\Sigma_f)/\Sigma_t$. At $k = 1$ the source bracket collapses to $c\Sigma_t\phi$ whatever the split between scattering and fission, so the critical size depends on c alone and the code takes the simplest split, $\Sigma_t = 1$, $\Sigma_s = 0$, $\nu\Sigma_f = c$. Question 5 has no such freedom and uses the tabulated cross sections directly.

3 Critical slab half-thickness by S_N

Reflective at $x = 0$, vacuum at $x = a/2$, 200 cells. The reference column is Assignment 1's exact-transport relation $a/2 = \frac{\pi}{2}\nu_0 - z_0$, evaluated from Case's tabulated ν_0 and z_0 .

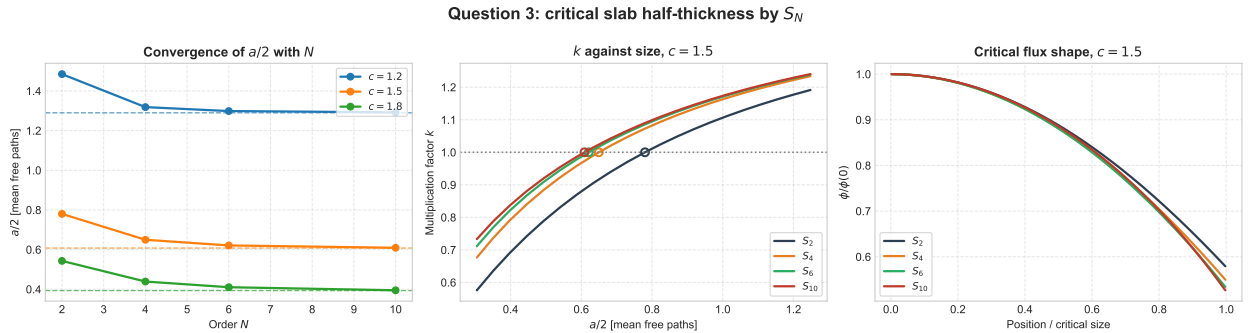


Figure 1: Left: convergence of $a/2$ with N , dashed lines the exact-transport values. Centre: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings. Right: the normalised critical flux, which barely moves with N — the order affects the boundary, not the interior.

The exact one-speed value at $c = 1.5$ is 0.605055 mfp, the Sood *et al.* Pu-239 slab (1.853722 cm at $\Sigma_t = 0.32640 \text{ cm}^{-1}$, the same table Assignment 1 Question 5 uses). The S_N sequence walks down onto it monotonically: 0.609042 at S_{10} , 0.606407 at S_{16} , 0.605631 at S_{24} , 0.605195 at S_{48} .

c	order	$a/2$ [mfp]	exact transport [mfp]	departure
1.2	S_2	1.48499	1.28981	+15.13%
1.2	S_4	1.31855	1.28981	+2.23%
1.2	S_6	1.29861	1.28981	+0.68%
1.2	S_{10}	1.29218	1.28981	+0.18%
1.5	S_2	0.78001	0.60762	+28.37%
1.5	S_4	0.64949	0.60762	+6.89%
1.5	S_6	0.62089	0.60762	+2.18%
1.5	S_{10}	0.60904	0.60762	+0.23%
1.8	S_2	0.54291	0.39314	+38.09%
1.8	S_4	0.43875	0.39314	+11.60%
1.8	S_6	0.41014	0.39314	+4.32%
1.8	S_{10}	0.39459	0.39314	+0.37%

Table 1: Critical half-thickness in mean free paths. The departure is entirely angular: the same radii are reproduced to seven digits on meshes from 50 to 800 cells.

P_N against S_{N+1}

In slab geometry the two are algebraically the same interior equations. Expanding ψ in $N + 1$ Legendre moments and truncating gives the same $N + 1$ coupled ODEs as collocating at the $N + 1$ Gauss–Legendre points, because a Gauss–Legendre rule of order $N + 1$ integrates polynomials up to degree $2N + 1$ exactly. **Any difference between P_N and S_{N+1} is therefore a difference of boundary condition**, and it is largest at low N where the boundary is the whole story:

- S_{N+1} sets $\psi(a/2, \mu_m) = 0$ on each incoming ordinate — the *Mark* condition.
- P_N needs conditions on moments instead. *Marshak*’s, $\int_0^1 P_{2j+1}(\mu)\psi d\mu = 0$, weight the whole incoming half-range; Mark’s pin the same isolated directions S_{N+1} does.

With Mark conditions the two agree; with Marshak they do not. Concretely at $c = 1.5$: S_2 above gives $a/2 = 0.78001$, and P_1 with the Mark extrapolation $l_0 = 1/\sqrt{3}$ gives $\arctan(1/Bl_0)/B = 0.78001$ — the same five digits — while P_1 with the Marshak $l_0 = 2/3$ gives 0.72348, a 7% spread between two methods that share every interior equation.

A second source of difference is that S_N is a collocation method: it satisfies the equation exactly along N directions and says nothing between them. At the vacuum face the leaked current $\sum_{\mu_m > 0} w_m \mu_m \psi_m$ is that of a flux strongly peaked towards $\mu = 1$, which a coarse quadrature underestimates — so the slab has to be made *thicker* than it should be to stay critical, and the table comes down from above.

4 Critical sphere radius by S_N

Reflective at $r = 0$, vacuum at $r = R$, 100 cells, with the angular redistribution term of (5). The reference is Assignment 1’s $\Sigma_t R_c = \pi\nu_0 - z_0$.

Cross-check between the two geometries. Both drive the asymptotic flux to zero at the same extrapolated surface, so $a/2 + z_0 = \frac{\pi}{2}\nu_0$ and $R_c + z_0 = \pi\nu_0$, and therefore

$$z_0 = R_c - 2(a/2) \tag{8}$$

c	order	R_c [mfp]	exact transport [mfp]	departure
1.2	S_2	3.06060	3.17204	−3.51%
1.2	S_4	3.14631	3.17204	−0.81%
1.2	S_6	3.15991	3.17204	−0.38%
1.2	S_{10}	3.16728	3.17204	−0.15%
1.5	S_2	1.60725	1.69010	−4.90%
1.5	S_4	1.66739	1.69010	−1.34%
1.5	S_6	1.67934	1.69010	−0.64%
1.5	S_{10}	1.68595	1.69010	−0.25%
1.8	S_2	1.11753	1.18295	−5.53%
1.8	S_4	1.16413	1.18295	−1.59%
1.8	S_6	1.17412	1.18295	−0.75%
1.8	S_{10}	1.17971	1.18295	−0.27%

Table 2: Critical radius in mean free paths. Note the sign: the sphere converges from *below*, the slab from above.

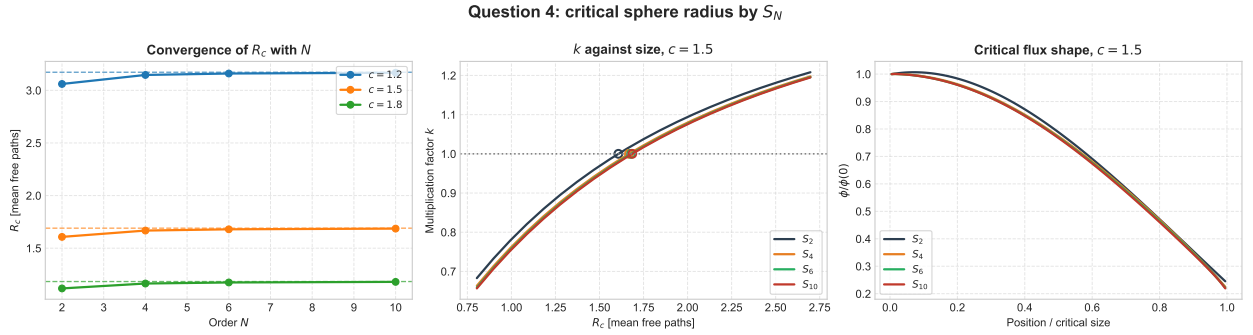


Figure 2: The same three panels for the sphere. The S_2 curve is the outlier in all three: in curvilinear geometry S_2 is not the P_1 equation, and the angular redistribution term it misrepresents is a loss from the interior directions.

which lets the two tables above check each other without any further calculation:

c	from S_2	from S_4	from S_6	from S_{10}	Case's table
1.2	0.091	0.509	0.563	0.583	0.592
1.5	0.047	0.368	0.438	0.468	0.475
1.8	0.032	0.287	0.354	0.391	0.397

Table 3: Extrapolation distance z_0 implied by the two geometries, in mean free paths. At S_{10} it is 1.5% below Case's value at all three c — the same shortfall three times, which is the relation's own asymptotic error. At S_2 it nearly vanishes, which is the sharpest statement of what low-order S_N gets wrong: the boundary, not the interior.

5 Critical mass of U-235 and Pu-239 by S_N

The cross sections are the Question 5 table of Assignment 1 — $\Sigma_t = 0.32640 \text{ cm}^{-1}$ for both, giving $c = 1.500$ for Pu-239 and $c = 1.300$ for U-235 — and the critical mass follows from the converged radius as $M_c = \frac{4}{3}\pi R_c^3 \rho$. Here $\Sigma_s > 0$, so the inner iteration does real work: about ten sweeps per outer, against one for the c -only problems above. The orders are S_2 to S_{16} rather than the S_2 – S_{10} of the previous questions, since the mass goes as the cube of the radius and is the more demanding quantity.

material	order	c	R_c [cm]	M_c [kg]	asympt. diffusion R_c [cm]	departure
Pu-239	S_2	1.500	4.9242	7.852	5.1890	−5.10%
Pu-239	S_4	1.500	5.1084	8.767	5.1890	−1.55%
Pu-239	S_8	1.500	5.1586	9.028	5.1890	−0.59%
Pu-239	S_{16}	1.500	5.1730	9.104	5.1890	−0.31%
U-235	S_2	1.300	7.1218	28.748	7.4339	−4.20%
U-235	S_4	1.300	7.3514	31.619	7.4339	−1.11%
U-235	S_8	1.300	7.4072	32.345	7.4339	−0.36%
U-235	S_{16}	1.300	7.4231	32.553	7.4339	−0.15%

Table 4: Critical radius and mass, against the asymptotic diffusion result of Assignment 1 Question 5. The mass is the cube of the radius, so the 5% radius error of S_2 is a 14% error in mass.

Diffusion overestimates both radii, and by more for Pu-239 than for U-235: the surface sits about two mean free paths from the centre in both cases, but the higher c makes the Pu-239 flux fall off faster and its surface angular flux more forward-peaked, which is exactly what a diffusion approximation cannot represent. S_{16} recovers the difference to within 0.3% of the transport-theory radius.

Question 5: criticality of the bare benchmark spheres by S_N

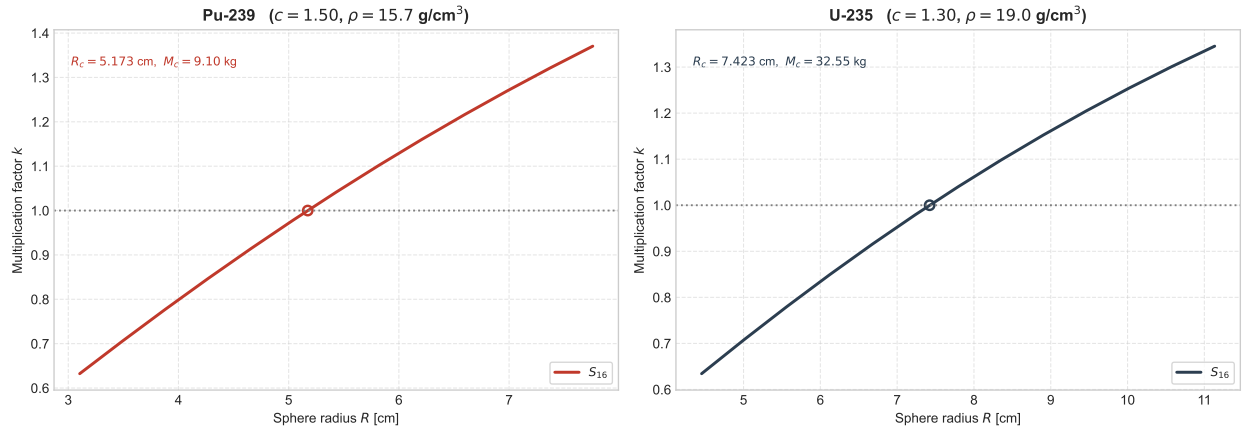


Figure 3: k against sphere radius at S_{16} , with the $k = 1$ crossing that fixes the critical mass.